

01/04/2025

① a) $\sqrt{1024} = 2^0$ b) $\sqrt[3]{\frac{1}{36}} = \frac{1}{6}$ c) $\sqrt[3]{900} = 3\sqrt[3]{10}$ d) $\sqrt[3]{64} = 4$ e) $\sqrt[3]{7964} \Rightarrow \sqrt[3]{2^2 \cdot 3^3 \cdot 7^2} = 42$ f) $\sqrt[3]{5^{-12}} \Rightarrow \sqrt[3]{\frac{1}{243}} = \frac{1}{3}$ g) $\sqrt[3]{\frac{1}{49}} = \frac{1}{7}$

h) $\sqrt[3]{729} \Rightarrow \sqrt[3]{3^6}$ i) $\sqrt[3]{-1331} \Rightarrow \sqrt[3]{(-11)^3} = -11$ j) $\sqrt[3]{-\frac{1}{32}} \rightarrow \sqrt[3]{-\frac{1}{2^5}} \rightarrow -\frac{1}{2}$ k) $\sqrt[3]{\frac{1}{27}} \rightarrow \frac{1}{3}$

② 6 //

③ $\sqrt[3]{\sqrt[3]{2}} \Rightarrow \left(\frac{1}{2}\right)^{\frac{1}{3}} \therefore \frac{1}{2} = \sqrt[3]{\frac{1}{2}}$ b) $\sqrt[3]{\sqrt[3]{\sqrt[3]{7}}} \Rightarrow \left(\left(\frac{1}{7}\right)^{\frac{1}{3}}\right)^{\frac{1}{3}} \Rightarrow \frac{1}{7}$ c) $\sqrt[3]{\sqrt[3]{\sqrt[3]{\sqrt[3]{23}}}} \Rightarrow \left(\left(\left(\left(\frac{1}{23}\right)^{\frac{1}{3}}\right)^{\frac{1}{3}}\right)^{\frac{1}{3}}\right)^{\frac{1}{3}} \rightarrow \sqrt[12]{23}$

④ a) $4^{\frac{9}{2}} = 2^9 = 2$ b) $2^{\frac{23}{2}} \rightarrow 2^9 = 16$ c) $32^{\frac{2}{3}} \rightarrow 32^{\frac{2}{3}} = 2$ d) $4^{\frac{9}{2}} \rightarrow 4^{\frac{9}{2}} \rightarrow (4^{\frac{3}{2}})^3 \rightarrow 2^9$ e) $8^{\frac{3}{2}} \rightarrow (8^{\frac{3}{2}})^2 \rightarrow 2 = 4$ f) $4^{\frac{9}{2}} \rightarrow (4^{\frac{3}{2}})^3 \rightarrow (2)^3 \rightarrow 8 //$

⑤ a) $\sqrt{8} \rightarrow 2\sqrt{2}$ b) $\sqrt{12} \rightarrow \sqrt{3 \cdot 4} \rightarrow 2\sqrt{3}$ c) $\sqrt{20} \rightarrow \sqrt{4 \cdot 5} \rightarrow 2\sqrt{5}$ d) $\sqrt{96} \rightarrow 4\sqrt{6}$ e) $\sqrt{192} \rightarrow 8\sqrt{3}$ f) $\sqrt{90} \rightarrow 3\sqrt{10}$

⑥ $\sqrt{18} + \sqrt{50} \rightarrow 3\sqrt{2} + 5\sqrt{2} \rightarrow 8\sqrt{2} //$ b) $\sqrt[3]{\frac{19}{125} + \sqrt{\frac{3}{5} - \frac{17}{25}}} \rightarrow \sqrt[3]{\frac{19}{125} + \frac{2}{5}} \rightarrow \sqrt[3]{\frac{69}{125}} \rightarrow \frac{4}{5}$

c) $\sqrt{\frac{2}{\sqrt[3]{2}}} \rightarrow \left(\frac{2}{2^{\frac{1}{3}}}\right)^{\frac{1}{2}} \rightarrow \left(2^{-\frac{1}{3}}\right)^{\frac{1}{2}} \rightarrow \left(2^{\frac{2}{3}}\right)^{\frac{1}{2}} \rightarrow 2^{\frac{1}{3}} \rightarrow \sqrt[3]{2} //$

d)

⑦ $\sqrt[16]{\frac{1}{2}} = \sqrt[4]{\frac{1}{2^4}} \rightarrow \frac{1}{2^{\frac{4}{4}}} = \frac{1}{2} \quad x=2 //$

⑧ $\sqrt{3}, \sqrt[3]{3}, \sqrt[4]{3} \rightarrow \frac{1}{3}, \frac{1}{9}, \frac{1}{27} \rightarrow \frac{1}{3} \cdot \frac{1}{9} \cdot \frac{1}{27} \rightarrow \frac{1}{72} \cdot \frac{13}{12} //$ * agora calculando com mesmo expoente:

⑨ $1 - \sqrt{2} < \sqrt{3} - 2 < 2 - \sqrt{5} < \sqrt{2} - 1$

⑩ $a = 4\sqrt[3]{5} \quad a \cdot b$
 $b = \sqrt[3]{375} = 5\sqrt[3]{3}$
 $\rightarrow 4 \cdot (4\sqrt[3]{5}) - 5\sqrt[3]{3} \Rightarrow 8\sqrt[3]{5} - 5\sqrt[3]{3} = 3\sqrt[3]{5}$

⑪ a) $\frac{2}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} \rightarrow \frac{2\sqrt{3}}{3}$ b) $\frac{(\sqrt{3}-1) \cdot (\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} \rightarrow \frac{4-2\sqrt{3}}{2} \rightarrow \frac{2(2-\sqrt{3})}{2} \rightarrow 2-\sqrt{3} //$ c) $\frac{\sqrt{2}}{(\sqrt{3}-\sqrt{2})} \cdot \frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}+\sqrt{2})} \rightarrow \frac{\sqrt{6}+2}{1} //$

d) $\frac{(\sqrt{3}+1) \cdot (\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} \rightarrow \frac{4+2\sqrt{3}}{2} \rightarrow 2+\sqrt{3} //$ e) $\frac{\sqrt{3}}{(\sqrt{2}-1)} \cdot \frac{(\sqrt{2}+1)}{(\sqrt{2}+1)} = \sqrt{3} \cdot (\sqrt{2}+1) //$
 $\hookrightarrow a^2 - b^2 = (a-b) \cdot (a+b)$

⑫ $\frac{7}{\sqrt{b}+\sqrt{b}} = \frac{\sqrt{b}-\sqrt{b}}{b-b}$

* Pode-se observar o padrão de cancelamento:

$7 + \sqrt{2} + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} \therefore \cancel{\sqrt{2}} + \cancel{\sqrt{3}} //$
 $\hookrightarrow 7 + 2 = 9$

E NÃO PODEMOS ESQUECER O -7

Primeiro termo:

$\frac{7}{1+\sqrt{2}} \cdot \frac{\sqrt{2}-1}{\sqrt{2}-1} \Rightarrow \frac{7(\sqrt{2}-1)}{\sqrt{2}-1+2-\sqrt{2}} \Rightarrow \frac{7(\sqrt{2}-1)}{1} //$ Ex - 7 não pode cancelar.

segundo termo

$\frac{1}{\sqrt{1}+\sqrt{5}} \cdot \frac{\sqrt{5}-\sqrt{1}}{\sqrt{5}-\sqrt{1}} \rightarrow \frac{\sqrt{5}-\sqrt{1}}{5-1} //$ * Segundo o lógico não resolve -7

⑬ $A^2 = 3 + \sqrt{5} + 3 - \sqrt{5} + 2 \cdot \frac{(\sqrt{3}+\sqrt{5}) \cdot (\sqrt{3}-\sqrt{5})}{\sqrt{4}} \rightarrow 6 + 4 = 10 \therefore A = \sqrt{10}$